

Project Design Phase-II

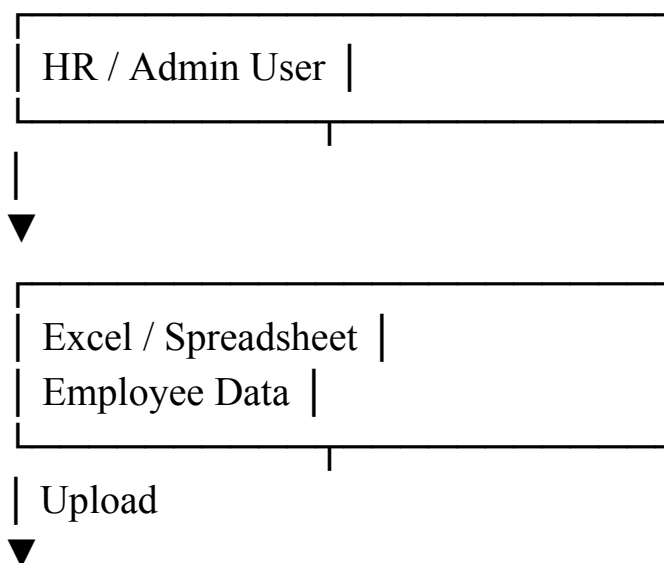
Technology Stack (Architecture & Stack)

Date	30 September 2026
Team ID	6ab24cc65bdbdc53c88cee
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Technical Architecture

The proposed solution uses the **ServiceNow platform** to import employee data from an Excel spreadsheet. The spreadsheet data is first loaded into an **Import Set table**. A **Transform Map** maps the source fields to the corresponding fields in the **Employee Test** target table. **Coalesce** is used to identify existing employee records and prevent duplicate records. The transformed data is then validated using **Transform History** and displayed through **Reports and Employee Analytics Dashboard**.

Architecture Flow



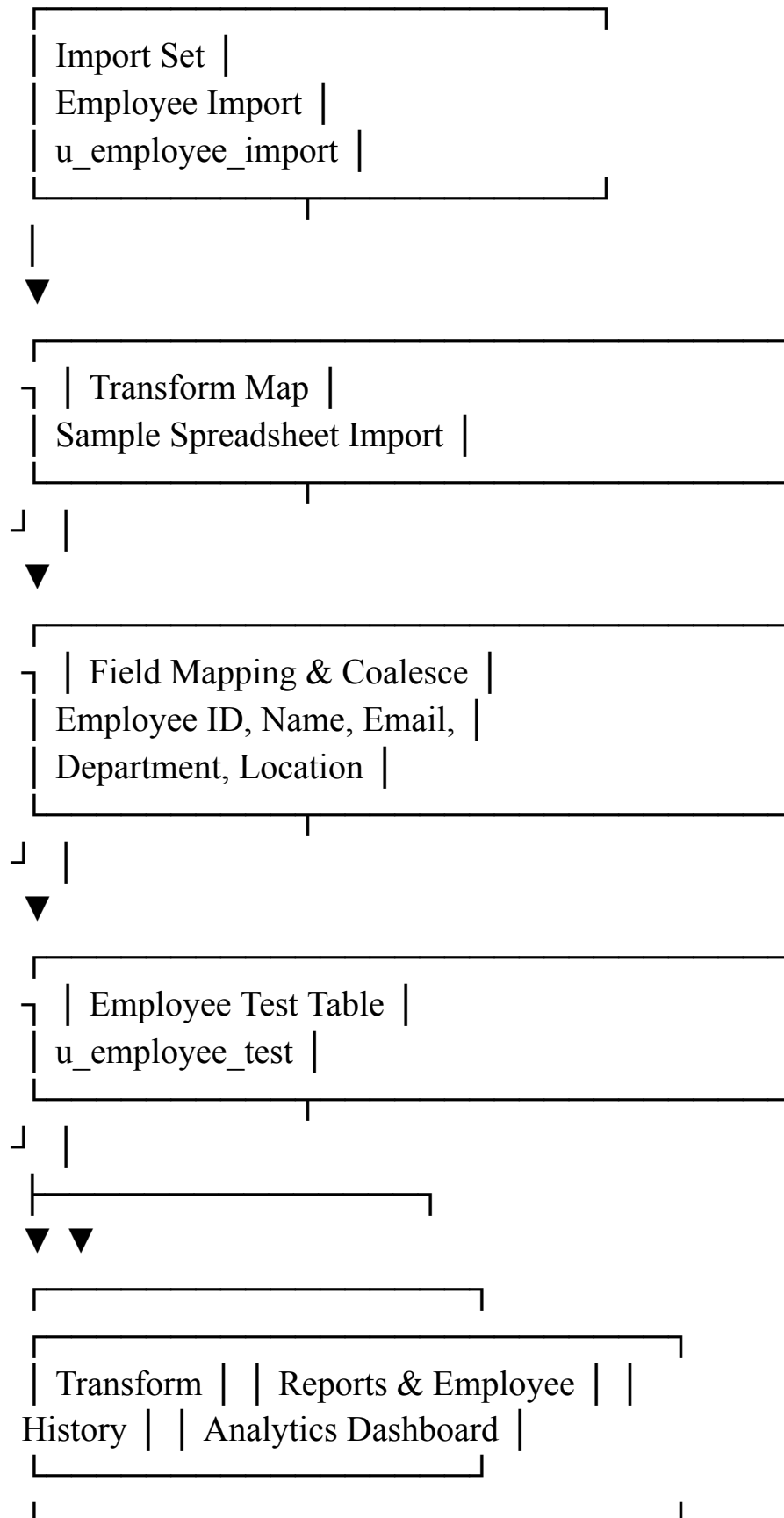


Table-1 : Components & Technologies

S.N o	Technology
Component Description	
1 User Interface	spreadsheet, manages duplicate employee imported data and views records. reports and dashboards. ServiceNow Web UI
2 Application Logic-1	Handles loading of employee data from the spreadsheet into the staging Import Set table. ServiceNow Import Sets
3 Application Logic-2	Transfers and transforms employee data from the Import Set Maps ServiceNow Transform
4 Application Logic-3	table to the Employee Test target table. Field Mapping
Interface through which the HR/Admin user uploads the employee	Maps employee fields and identifies existing records to prevent Coalesce
5 Database	Stores the imported employee data and transformed employee records in ServiceNow tables. ServiceNow Database
6 Cloud Database	Not separately used in the proposed solution. within the ServiceNow platform. Employee data is stored ServiceNow Platform
7 File Storage	Stores the prepared employee spreadsheet file used for importing Excel / Spreadsheet File
8 External API-1	employee spreadsheet data. Not Applicable
No external API is required for importing and transforming	9 External API-2 10 Machine

Learning Model	No machine learning model is required. The solution uses	cloud environment. Not Applicable
11 Infrastructure (Server / Cloud)	rule-based data import, field mapping and transformation.	Not Applicable
No external API is required for employee data transformation and validation.	The application is developed and executed on the ServiceNow platform using its web-based	ServiceNow Cloud Platform

Table-2: Application Characteristics

S.No	framework is required.	
Characteristics	The solution uses the built-in	
Description	ServiceNow platform features for data import, transformation, reporting and dashboard creation.	ServiceNow Roles and Access Controls
1 Open-Source Frameworks		3 Scalable
	Access to employee data and application features is controlled through ServiceNow user roles and access controls. Employee data is managed within the ServiceNow platform.	Architecture
2 Security Implementations	ServiceNow Platform	The solution can handle increasing employee records by using Import Sets and Transform Maps for repeated spreadsheet imports. The target table can store additional employee records as
No separate open-source		

required.
ServiceNow Import

Sets and

Transform Maps

4 Availability The solution is available through the ServiceNow web platform whenever the ServiceNow instance is accessible. Users can perform imports, validation and reporting through the platform.

5 Performance Import Sets provide a structured staging process, while Transform Maps automate field mapping and data transformation. Coalesce helps avoid unnecessary duplicate records during repeated imports.
ServiceNow Cloud
Platform

Import Sets, Transform Maps and
Coalesce